MIL-T-60876A(AR)
30 October 1985
SUPERSEDING
MIL-T-60876(MU)
15 August 1969

MILITARY SPECIFICATION

TEST SET, SERVO AMPLIFIER, TS-2658/TSM-100A

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of Servo Amplifier Test Set, hereinafter referred to as the "test set" which forms a part of the Electronics System Test Set type AN/TSM-100A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.2 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-T-43435	-	Tape: Impregnated, Lacing and Tying
MIL-C-46357	-	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4931

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

MIL-C-60898 - Circuit Card Assemblies, General Specification for
MIL-C-60898/11 - Circuit Card Assembly, Servo Amplifier

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-810 - Environmental Test Methods

2.1.2 Other government documents, drawings and publications. The following other Government drawings form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

11748029 - Panel Assembly, Servo Amplifier Test Unit
12000196 - Schematic Diagram, Electrical, Servo Amplifier Test Unit

PACKAGING DATA SHEET

11748029 - Packaging of Test Set, Servo Amplifier

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Fabrication. The test set shall be manufactured in accordance with Drawing 11748029 and drawings pertaining thereto. The subassemblies shall have been inspected and accepted in accordance with their respective drawings or specifications prior to their installation into the test set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

3.2 General specifications. The test set shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

c. Part identification and marking.

d. Workmanship.

e. Soldering electrical connections.

3.2.2 MIL-C-46357.

a. Etched circuit assembly. The requirements for etched circuit assemblies of MIL-C-46357 shall apply with modification as follows:

(1) Plated etched circuit boards. In addition, assemblies shall not contain copper or nickel visible through hot solder dip, when viewed at 10 times magnification.

(2) Location grid. The component mounting holes shall be within 0.007 inch of the true grid intersection, instead of 0.005 inch.

(3) Transistor and diode leads. Transistor and metal cased diode leads shall be covered with sleeving in areas where there is a possibility of shorting. No color coding of sleeving shall be required.

(4) Drying. The drying process shall not be required.

b. Wiring. The requirements for wiring of MIL-C-46357 shall apply with modification as follows:

(1) Spot ties. Tape, Type 1, finish B per MIL-T-43435 may be utilized without the need for varnish on the knots.

(2) Wire termination. Removal of insulation by controlled heat methods shall be acceptable.

(3) Chassis cleaning. Compressed air or vacuum may be utilized for cleaning. Vibration shall not be used to clean the chassis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

c. Threaded fasteners. The requirements for threaded fasteners of MIL-C-46357 shall apply with modification as follows:

(1) Thread projection. Maximum projection of screws shall be that of the next standard length screw beyond the minimum projection, unless electrical or mechanical interference exists.

d. Marking. The requirements for marking of MIL-C-46357 shall apply with modification as follows:

(1) Etching and engraving. Solvo photo etch process is acceptable.

3.2.3 Circuit card assembly. The circuit card assembly shall meet the requirements of MIL-C 60898/11.

3.3 Environmental service conditions.

3.3.1 Vibration. The test set shall meet the requirements of 3.4 and 3.5 after being subjected to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range (Hz ± 2 Percent)	Excursion (double amplitude) (inches ± 10 percent)
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle per second per second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. The test set shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating. The test set shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

3.3.3.2 Non-operating. The test set shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure enountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The test set shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Non-operating (storage). The test set shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 48 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The test set shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.

3.4.1 Continuity. The circuitry shall be in accordance with electrical schematic drawing 12000196. Each conductor shall be electrically continuous as specified on the schematic diagram. Switches shall be tested in all position and for each switch function in accordance with the wiring runs list 11748029. The wiring runs list shall take precedence over the schematic. The symbols referenced in subsequent paragraphs pertain to this schematic and are used therein.

3.5 Performance.

3.5.1 Feedback circuit.

3.5.1.1 CCW input. With 70.0 ± 0.5 vdc applied between J115-C(-) and J115-D(+), the output voltage between TP2 and GROUND (TP4) shall be as specified in Table I for the corresponding positions of the TEST SELECT switch, HAND CONTROL knob or DEAD BAND knob.

3.5.1.2 CW input. With 70.0 ± 0.5 vdc applied between J115-C(+) and J115-D(-), the output voltage between TP1 and GROUND (TP4) shall be as specified in Table I for the corresponding positions of the TEST SELECT switch, HAND CONTROL knob or DEAD BAND knob except that CCW and CW rotations are reversed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

TABLE I. Voltage output test.

TEST SELECT Switch Position	HAND CONTROL Knob Position	DEAD BAND Knob Position	OUTPUT VOLTAGE (Vdc)
RADAR MODE	MAX CCW (See Note 1)		-3.20 to -3.55
RADAR MODE	MAX CW		-0.72 to -0.78
DEAD BAND		MAX CCW	-2.08 to -2.25
DEAD BAND		MAX CW	-1.88 to -2.02
GRD MODE	MAX CCW		-2.17 to -2.54
GRD MODE	MAX CW		-0.50 to -0.56
MAX SLEW			-1.96 to -2.19
MAX SLEW			See Note 2 (0 to +.035)

NOTES:

1. Maximum CCW or CW position for HAND CONTROL knob shall be the position of the knob just before a discontinuity is observed in the output voltage at TP2 when the knob is rotated in the counterclockwise or clockwise direction respectively. This position shall be 175 ± 5 degrees from the 0 position.

2. Output voltage of 0 to +.035 vdc shall be present between TP1 and GROUND (TP4) when the applied voltage is connected as specified in 3.5.1.1. When the applied voltage is connected as specified in 3.5.1.2, an output voltage of 0 to .035 vdc shall be present between TP2 and GROUND (TP4).

3.5.2 Switch functions.

3.5.2.1 DIGITAL VM SELECT switch. The resistance between J25 and the pins specified in Table II shall be less than 0.5 ohms when the DIGITAL VM SELECT switch is in the corresponding position specified. In any other switch position, the resistance shall be greater than 1 megohm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

TABLE II. Resistance test.

DIGITAL VM SELECT Switch Position	Connector and Pin
+15 V	J20-Z
-15 V	J20-A
ERROR SIGNAL	J20-G
SLAVE DRIVE SIGNAL	J21-C
CCW OUPT	J21-K
CW OUPT	J21-L
MOTOR CURRENT	J20-F
PWM INPUT	J20-H

3.5.2.2 SCOPE SELECT switch. The resistance between J26 and the pins specified in Table III shall be less than 0.5 ohms when the SCOPE SELECT switch is in the corresponding position specified. In any other switch position, the resistance shall be greater than 1 megohm.

TABLE III. Resistance test.

SCOPE SELECT Switch Position	Connector and Pin
OSCILLATOR	J20-V
ERROR SIGNAL	J20-G
PWM INPUT	J20-H
CW BASE DRIVE	J20-C
CCW BASE DRIVE	J20-B
MOTOR CURRENT	J20-F
CCW POWER MODULE OUPT	J20-T
CW POWER MODULE OUPT	J20-Y
CCW OUPT	J21-K
CW OUPT	J21-L

3.5.2.3 Continuity.

3.5.2.3.1 GROUND continuity. There shall be continuity between GROUND (TP4) and the connector pins as listed in Table IV. The resistance between GROUND and these pins shall be less than 0.5 ohms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

TABLE IV. Resistance test.

Connector and Pin	Connector and Pin
J115-A	J21-E
J115-F	J21-F
J115-J	J21-H
J115-N	J21-J
J115-T	J21-T
J115-W	J21-W
J115-Y	

3.5.2.3.2 TP Continuity. The resistance between the Test Points (TPs) and the connector pins, listed in Table V, shall be less than 0.5 ohms.

TABLE V. Resistance test.

Front Panel Test Point	Connector Pin
TP1	J21-G
TP2	J21-R
TP3	J25
GROUND (TP4)	J115-F
SERVO LOAD + (TP5)	J21-M
SERVO LOAD - (TP6)	J21-M
TACH + (TP7)	J115-C
TACH - (TP8)	J115-D

3.5.2.4 ON/OFF and LOAD SELECT switches. The servo amplifier panel shall meet the requirements of Table VI, when connected as shown in Figure 1.

TABLE VI. Switch position test.

ON/OFF Switch	LOAD SELECT Switch Position	+24 vdc Lamp	Load On Lamp	Voltage Between Connector Pin and GROUND (TP4) (Tol. ± 0.2 Vdc)
OFF	OFF	OFF	OFF	J21-M, J115-G, J115-H, J115-U, J115-V, J115-X → 0vdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

TABLE VI. Switch position test.-(continued).

ON/OFF Switch	LOAD SELECT Switch Position	+24 vdc Lamp	Load On Lamp	Voltage Between Connector Pin and GROUND (TP4) (Tol). ± 0.2 Vdc)
ON	OFF	ON	OFF	J21-M 23.9vdc J115-G 0vdc J115-H 0vdc J115-U 23.9vdc J115-V 0vdc J115-X 0vdc
ON	1/3	ON	ON	J21-M, J115-G, J115-H, J115-U 23.9vdc J115-V 0vdc J115-X 0vdc
ON	2/3	ON	ON	J21-M, J115-G, J115-H, J115-U, J115-X 23.9vdc J115-V 0vdc
ON	FULL	ON	ON	J21-M, J115-G, J115-H, J115-U, J115-V, J115-X 23.9vdc

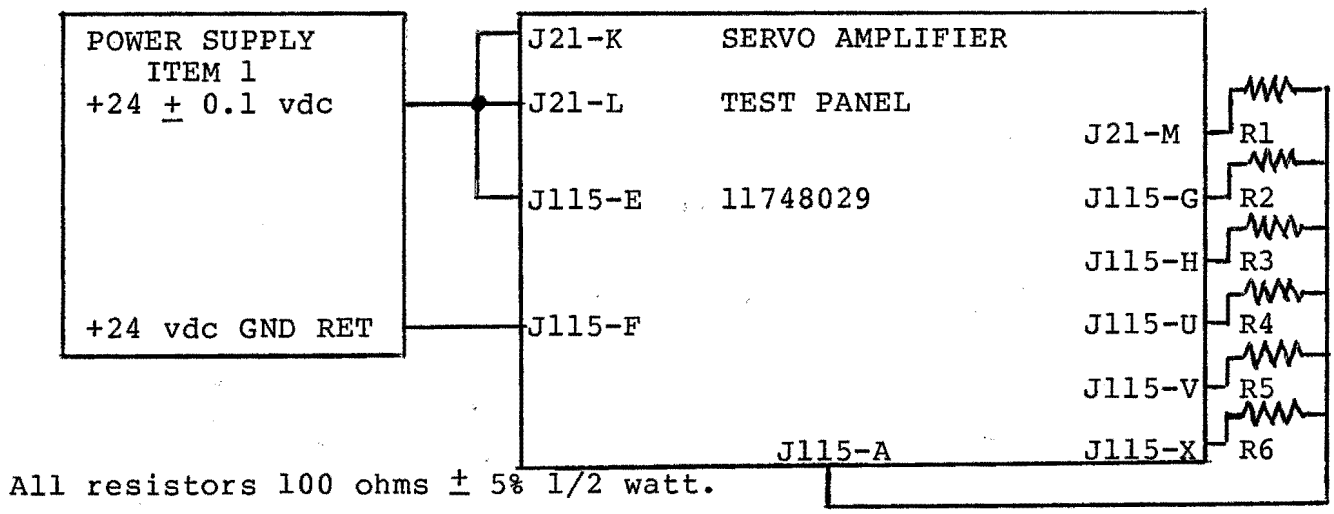


FIGURE 1. Test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

3.5.2.5 Circuit breakers.

3.5.2.5.1 7.5 Amperes. The 7.5 amp circuit breaker shall not open when subject to a current of 7.5 amperes. At 200% over load (15 amperes) the circuit breakers shall open within 60 seconds.

3.5.2.5.2 35 Amperes. The 35 amp circuit breaker shall not open when subjected to 35 amperes. At 200% over load (70 ampere) the circuit breaker shall open within 150 seconds.

3.6 First article. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.7 Workmanship. The requirements of MIL-STD-454 and other applicable specifications shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Unless otherwise specified in the item detail specification, or contract, the following types of inspection shall be conducted on the unit or product:

a. First article inspection (4.3)

b. Quality conformance inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting officer (see 6.2) for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 1 test set.

4.3.2 Inspection to be performed. First article sample may be subjected by the Government to any or all of the examinations and tests specified in 4.4.2 of this specification and to any or all requirements of the applicable drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

4.3.3 Rejection. If any assembly, component or test specimen of the test set fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination and tests. Except where otherwise specified herein, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables VII and VIII and methods of inspection in 4.6. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. Workmanship shall be in accordance with MIL-STD-454 and other applicable specifications.

4.4.2.1 Components and subassembies. Printed circuit subassemblies shall be inspected in accordance with the inspection provisions of MIL-C-60898. All other components and subassemblies shall be inspected to the applicable requirements of MIL-F-13926.

4.4.2.2 Control tests. One test set shall be selected at random as a control sample from each 100 produced. The control samples shall meet the requirements and tests of Table VII after they have met the requirements and tests of Table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

TABLE VII. Control Test Sampling.

Test	Requirement	Test Procedures
Fabrication	3.1	Applicable/Drawings Visual
General Specifications	3.2	Visual/Tactile
Vibration	3.3.1	4.6.1.1
Shock	3.3.2	4.6.1.2
Altitude	3.3.3	4.6.1.3
Temperature	3.3.4	4.6.1.4
Humidity	3.3.5	4.6.1.5

4.4.3 Rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4.4 Functional tests. The requirements and tests in Table VIII shall be inspected on a 100 percent basis. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions.

a. Temperature +50° to +90°F

b. Relative Humidity 90% RH max.

c. Pressure 28 to 32 inches of Hg

TABLE VIII. Functional Test.
(100 percent inspection)

Characteristic	Requirement	Procedure
100 Continuity	3.4.1/3.5.2.3	4.6.2.1/4.7.3
101 CCW input	3.5.1.1	4.7.1.1
102 CW input	3.5.1.2	4.7.1.2
Switch functions		
103 Digital VM select sw	3.5.2.1	4.7.2.1
104 Scope select sw	3.5.2.2	4.7.2.2
105 On/Off and load select sw	3.5.2.4	4.7.3.3
Circuit breakers		
106 7.5 amperes	3.5.2.5.1	4.7.4.1
107 35 amperes	3.5.2.5.2	4.7.4.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

4.5 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the referenced specifications and standards cited in 4.4.2. Inspection equipment shall be as listed in 4.5.1. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See 6.3 of this specification.

4.5.1 Inspection equipment. The following inspection equipment or equivalent shall be used in the performance of the applicable test as specified in 4.6.

TABLE IX. Test equipment list

Item number	Description
1. Power Supply	0-30 vdc adjustable,Load regulation of ±0.5% or less. Current; 6 amperes minimum. Less than 0.5% ripple.
2. Resistor (R1 thru R6)	100 ohms, 5% tolerance, 1/2 W.
3. Power Supply	70.0 ± 0.5 vdc output. Load regulation of ± 0.5%. Current; 250 ma minimum. Less than 0.05% ripple +, -, and ground connections required.
4. Ohmmeter	To measure resistance, zero ohm to greater than 1 megohm, min. of 1% accuracy.
5. Digital Voltmeter	Auto scaling. Resolution and accuracy of 1% or better from 1 volt to 100 volts.
6. Power Supply	D.C. power supply and an ammeter indicating load current and an adjustable current limiting control. The capacity will be 70 amperes or greater.

4.6 Methods of inspection.

4.6.1 Environmental. Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

4.6.1.1 Vibration. The test set shall be mounted on the vibration table in normal operating position. The test shall be performed as specified in Method 514, Procedure I, Category 1, of MIL-STD-810 except as modified by paragraph 3.3.1 herein. After vibration, the test set shall be tested as given in 4.6.2 and 4.7 for compliance with 3.3.1.

4.6.1.2 Shock. This test shall be performed as specified in Procedure I Method 516 of MIL-STD-810 except that shock pulses shall be as specified in 3.3.2. After the shock impulses, the test set shall be tested as specified in 4.6.2 and 4.7 for compliance with 3.3.2.

4.6.1.3 Altitude (nonoperating and operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.6.2 and 4.7 for compliance with 3.3.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). This test shall be performed as specified in Procedure I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.6.2 and 4.7 for compliance with 3.3.4.

4.6.1.4.2 Low (operating and nonoperating). This test shall be performed as specified in Procedure I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.6.2 and 4.7 for compliance with 3.3.4.

4.6.1.5 Humidity (nonoperating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7 the operational checkout shall be performed at the end of the tenth cycle only at standard ambient conditions.

4.6.2 Electrical.

4.6.2.1 Continuity. A continuity test shall be performed using an ohmmeter to verify conformance of 3.4.

4.7 Performance.

4.7.1 Feedback circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

4.7.1.1 CCW input. Connect the 70.0 ± 0.5 vdc power supply, between J115-D (+) and J115-C (-). Do not reference the (+) or (-) to ground. Referring to Table I, set the TEST SELECT switch, HAND CONTROL and DEAD BAND controls as specified for each line of the table. Verify that the measured output voltages between TP2 and GROUND conform to the requirements of 3.5.1.1. After test remove power supply.

4.7.1.2 CW input. Connect the 70.0 ± 0.5 vdc power supply, between J115-C (+) and J115-D (-). Do not reference the (+) or (-) to ground. Referring to Table I, set the TEST SELECT switch, HAND CONTROL and DEAD BAND controls as specified for each line of the table. Verify that the measured output voltages between TP1 and GROUND conform to the requirements of 3.5.1.2. After test remove power supply.

4.7.2 Switch functions.

4.7.2.1 DIGITAL VM SELECT switch. Using an ohmmeter, measure the resistance between the center pin of J25 and each connector pin with the DIGITAL VM SELECT switch in the corresponding positions as specified in Table II. Verify that the measured resistance conforms to requirements of 3.5.2.1 for each specified switch position.

4.7.2.2 SCOPE SELECT switch. Using an ohmmeter, measure the resistance between the center pin of J26 and each connector pin with the SCOPE SELECT switch in the corresponding positions as specified in Table III. Verify that the measured resistance conforms to requirements of 3.5.2.2 for each switch position.

4.7.3 Continuity.

4.7.3.1 GROUND continuity. With an ohmmeter, measure the resistance between GROUND (TP4) and the connector pins as outlined in Table IV of 3.5.2.3.1. The resistance measured between GROUND and each connector pin shall be less than 0.5 ohms.

4.7.3.2 TP continuity. With an ohmmeter, measure the resistance between the Front Panel Test Points and associated connector pins as outlined in Table V of 3.5.2.3.2. The resistance measured between each test point and associated connector pin shall be less than 0.5 ohms.

4.7.3.3 ON/OFF and LOAD SELECT switches. With the Servo Amplifier test panel connected as shown in Figure 1, measure the voltages between the connector pins and ground and observe the front panel lamps when the associate switches are positioned as outlined in Table VI of paragraph 3.5.2.4. After test remove +24 vdc supply and all resistors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

4.7.4 Circuit breakers.

4.7.4.1 7.5 amperes. Connect a current source across the terminals of the 7.5 amperes circuit breaker. With 7.5 amperes applied, the circuit breaker shall not trip. Test for a minimum of 15 minutes. Increase current to 15 amperes. Verify that the circuit breaker shall open within 60 seconds and that it conforms to the requirement of 3.5.2.5.1. After test, remove current source and reset circuit breaker.

4.7.4.2 35 amperes. Connect a current source across the terminals of the 35 amperes circuit breaker. With 35 amperes applied, the circuit breaker shall not trip. Test for a minimum of 15 minutes. Increase current to 70 amperes. Verify that the circuit breaker shall open within 150 seconds and that it conforms to the requirement of 3.5.2.5.2. After test, remove current source and reset circuit breaker.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet 11748029.

6. NOTES

6.1 Intended use. The Test Set is designed for rack-mounting into the AN/TSM-100A Electronics System Test Set, which is used for troubleshooting and performance checking of specific weapon system components.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Applicable stock number.

c. A block of serial numbers to identify each test set.

d. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

e. First Article Sample requirements, if other than specified in Section 4 of item detail specification.

f. Packaging requirements, if other than specified in Section 5 of item detail specification.

g. Certificate of conformance for each lot or shipment or product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

6.2.2 Contract data requirements.

a. Contract data requirements for inspection equipment designs conforming to Data Item Description DI-R-1714, with the appropriate addendum to each. See 6.3

b. Data Cards. When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410.

6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government designs and contractor designs. Government designs are designated by drawing numbers in the appropriate paragraph in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government design is provided; however, unless otherwise specified, the contractor may submit alternate or modified contractor designs in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 Government designs. Government designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in DOD-STD-100.

c. An envelope drawing, as defiend in DOD-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for ALL inspection equipment for which Government designs are not specified and include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with Government designs if not otherwise covered thereby (see 6.3.1c). Designs

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

shall be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements List (DD Form 1423). The item detail specification number, paragraph number and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, timeframe and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAF-I(D), Dover, NJ 07801-5001.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as US Army Armament Research and Development Command (AMCCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A215)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.